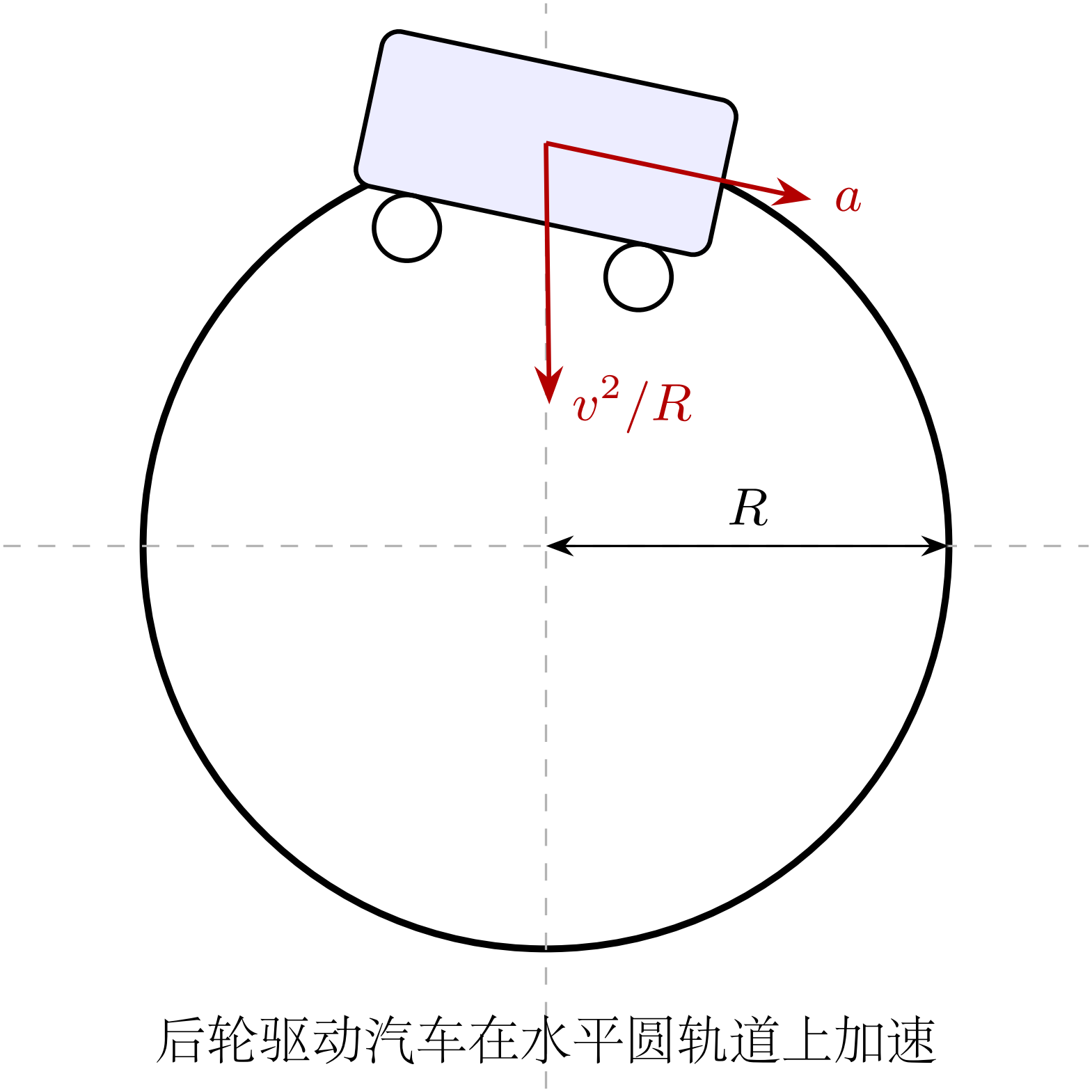
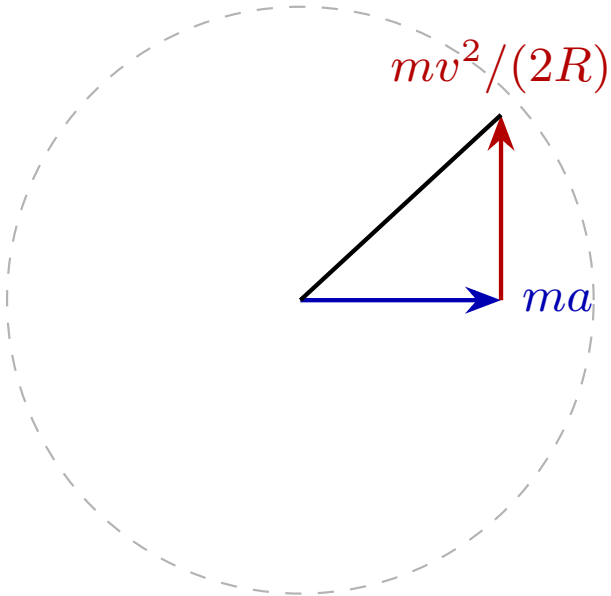


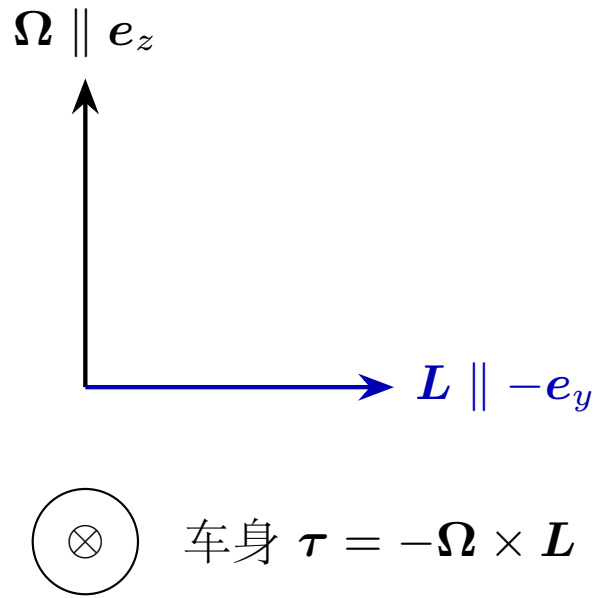
后轮驱动汽车在水平圆轨道上加速



驱动后轮摩擦圆

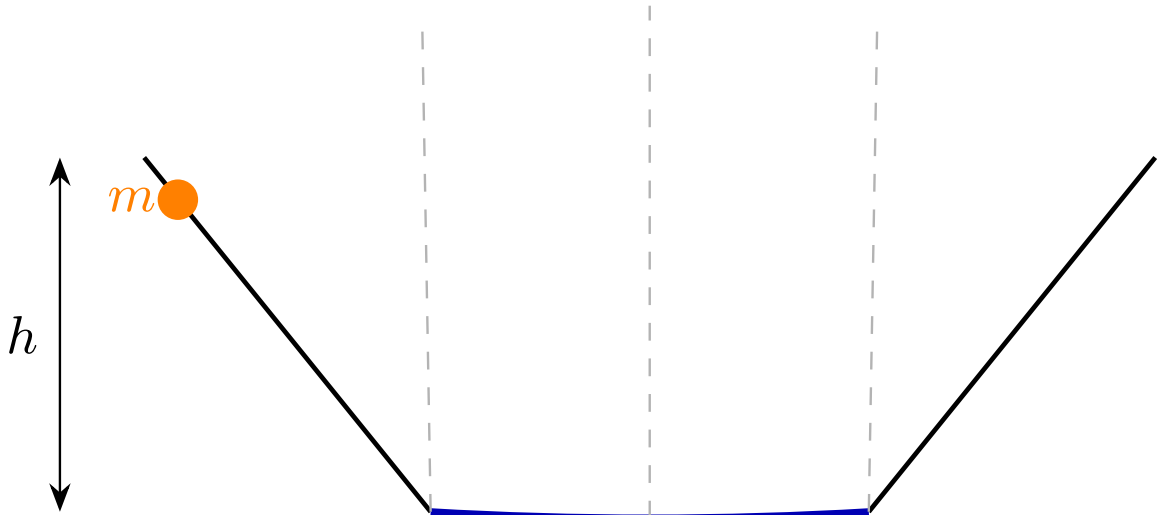
$$\sqrt{(ma)^2+[mv^2/(2R)]^2}\le \mu mg/2$$

横向等分另需忽略偏航惯性



沿车头看截面，$\boldsymbol{e}_x$ 朝读者

车轮受 $+\boldsymbol{e}_x$ 力矩，车身受反向力矩



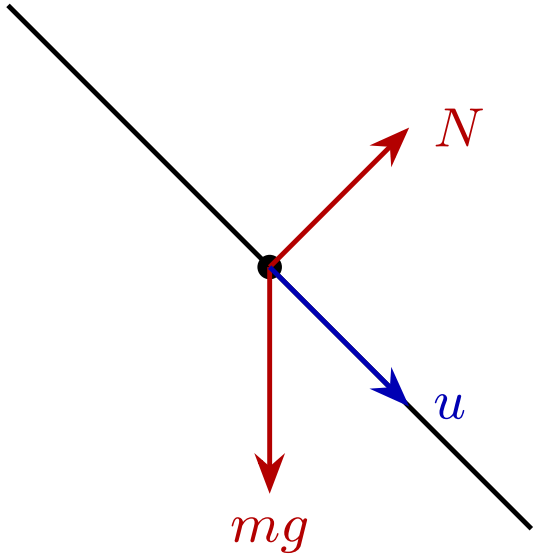
斜面 $\alpha = \pi/4$

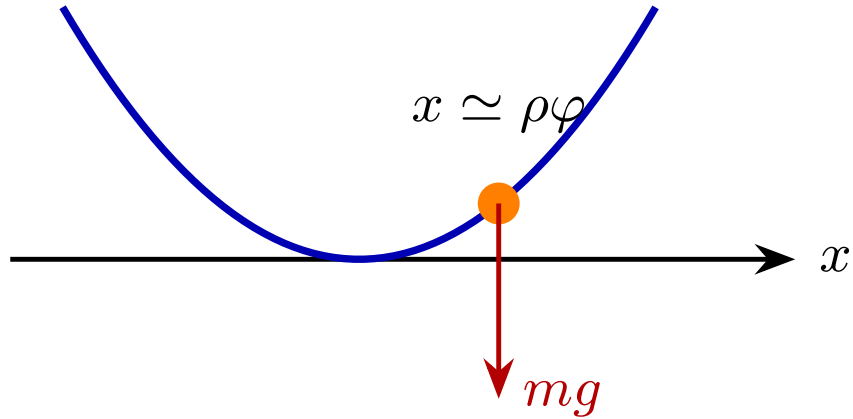
圆弧半角 $\theta = 0.01$

$\rho = 10\,\mathrm{m}$

两段不相切，原几何示意

不能直接当连续光滑轨道





斜面局部有效质量

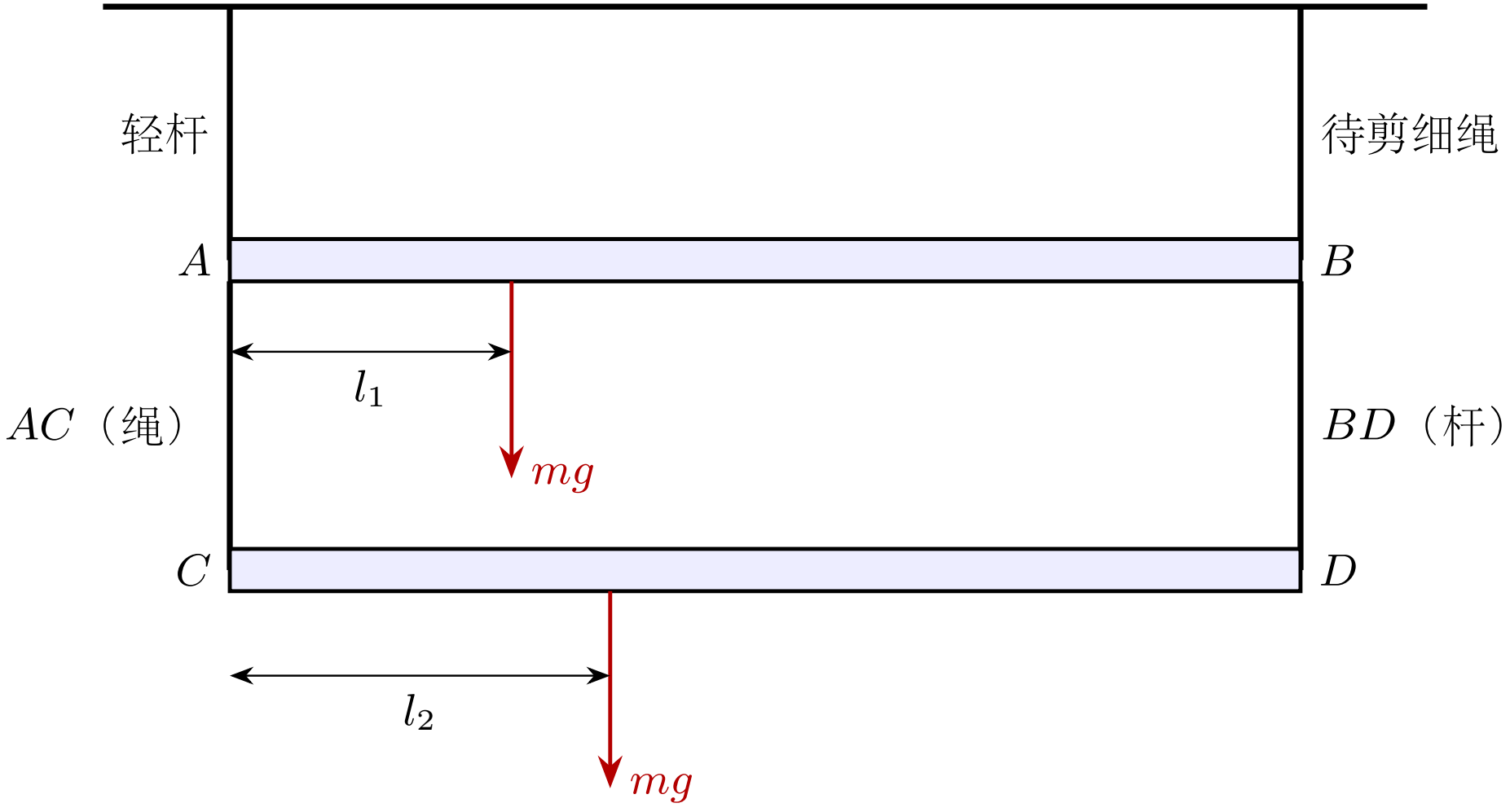
$m_s = m(M + m\cos^2\alpha)/(M + m)$

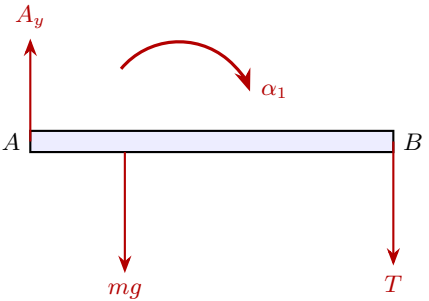
圆弧局部有效质量

$m_c = mM/(M + m)$

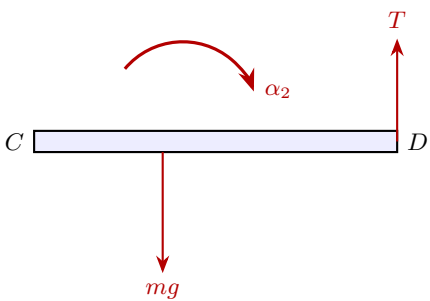
两个局部模型；过渡另需规则

无损短过渡：$m_s u_1^2 = m_c u_c^2$；进入圆弧使用 u_c

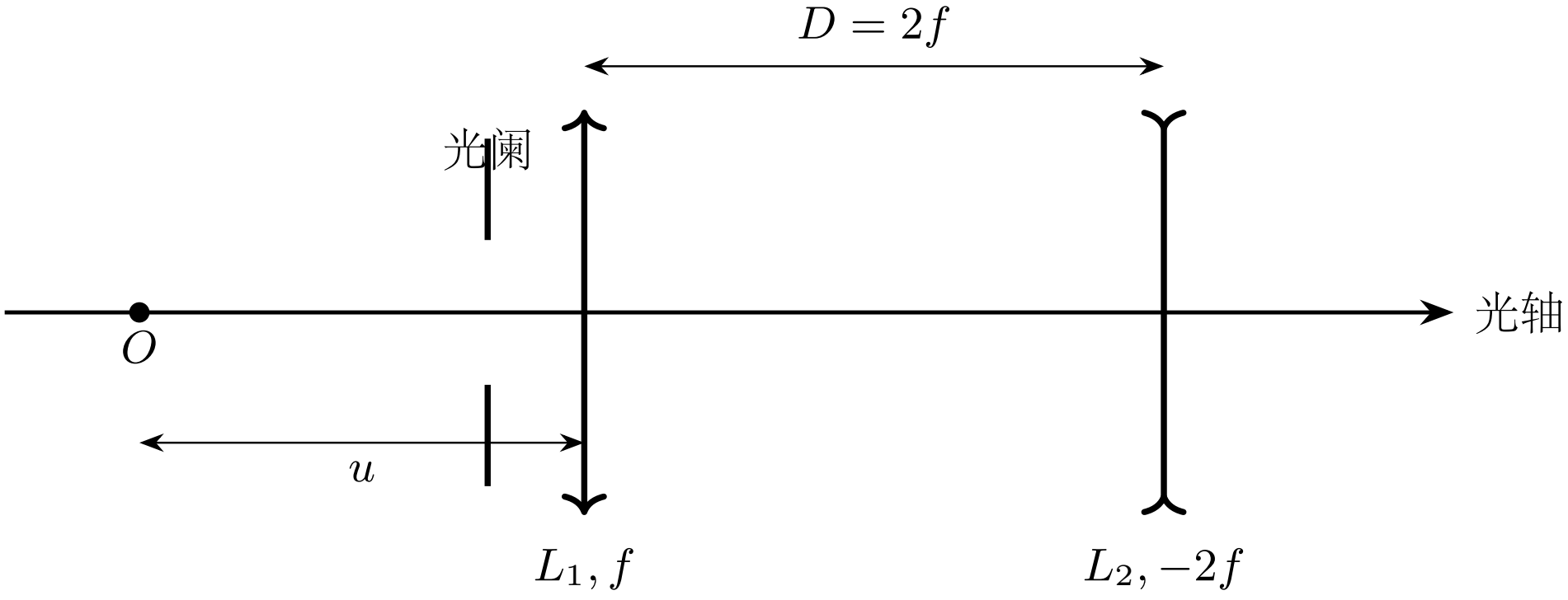

轻杆
待剪细绳
A
B
AC（绳）
BD（杆）
C
D
l_1
mg
l_2
mg

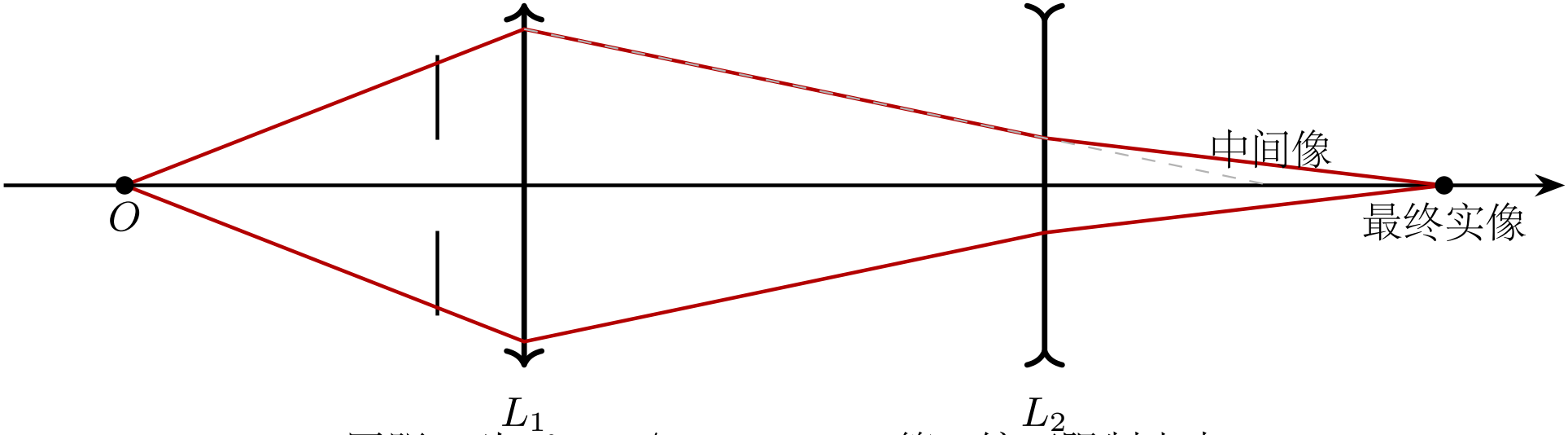


$m(a^2 + l_1^2)\alpha_1 = mgl_1 + Tl$



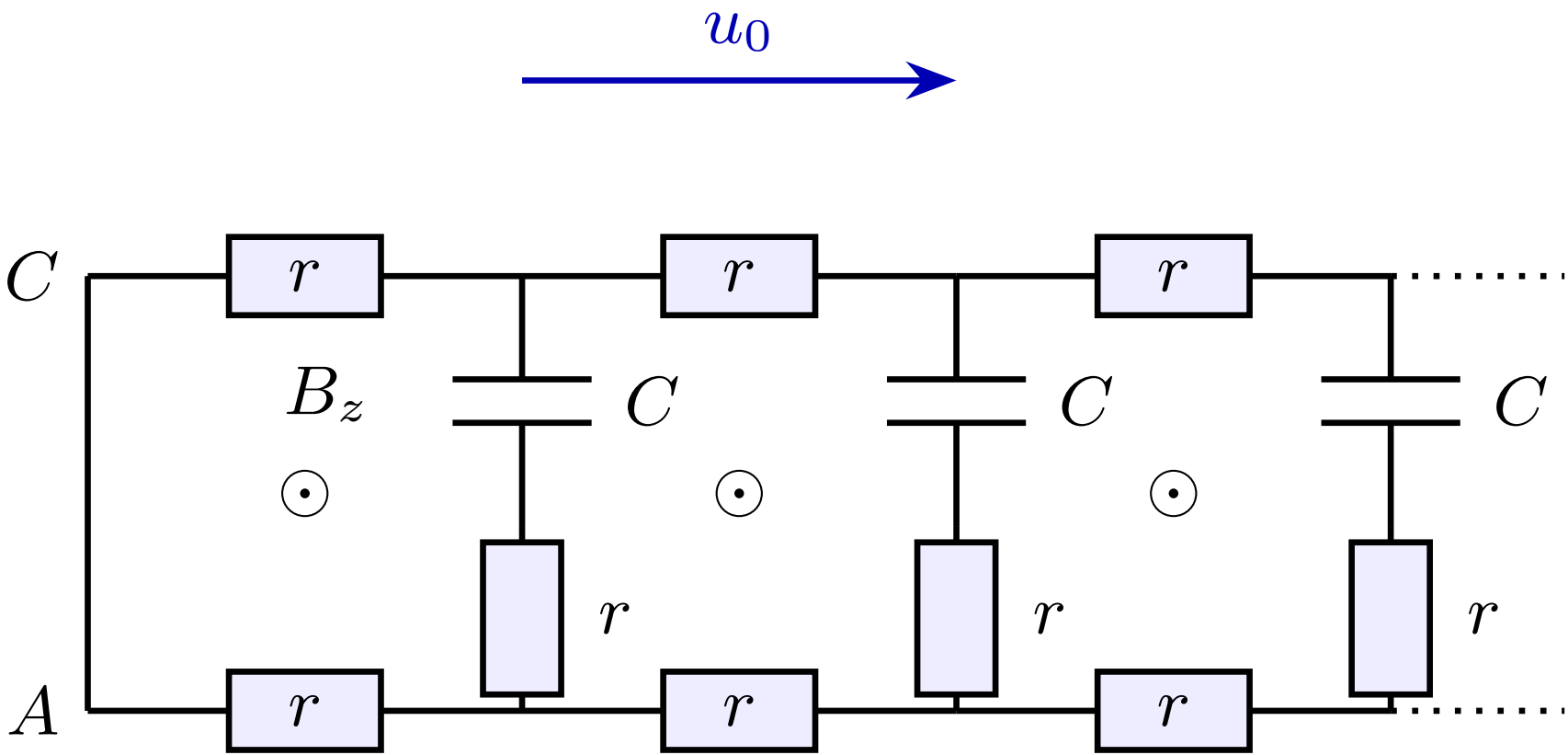
约束：$a_{D,y} = a_{B,y}$


$D = 2f$
光阑
光轴
O
u
L_1, f
$L_2, -2f$



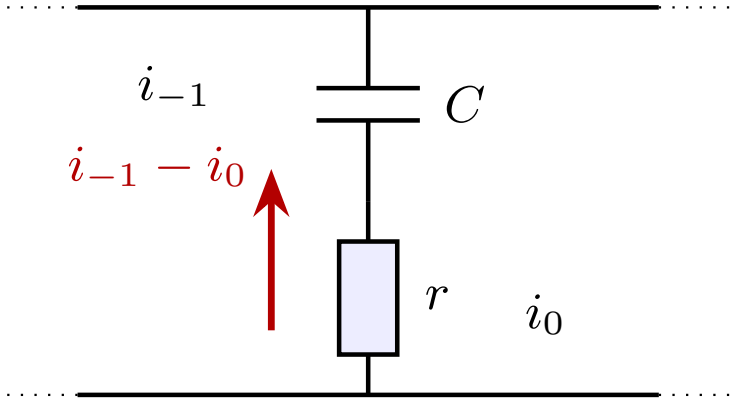
图距 2 为 f；$y_2/y_1 = 0.303$，第二镜不限制光束

光阑点源距离 $d = u_3 - f/3$，亮度比取立体角



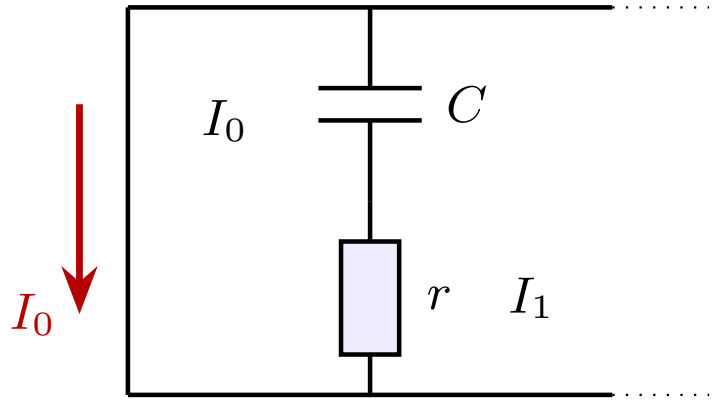
原图端部：AC 为导线，向右延伸

两种边界不可使用同一支路电流公式



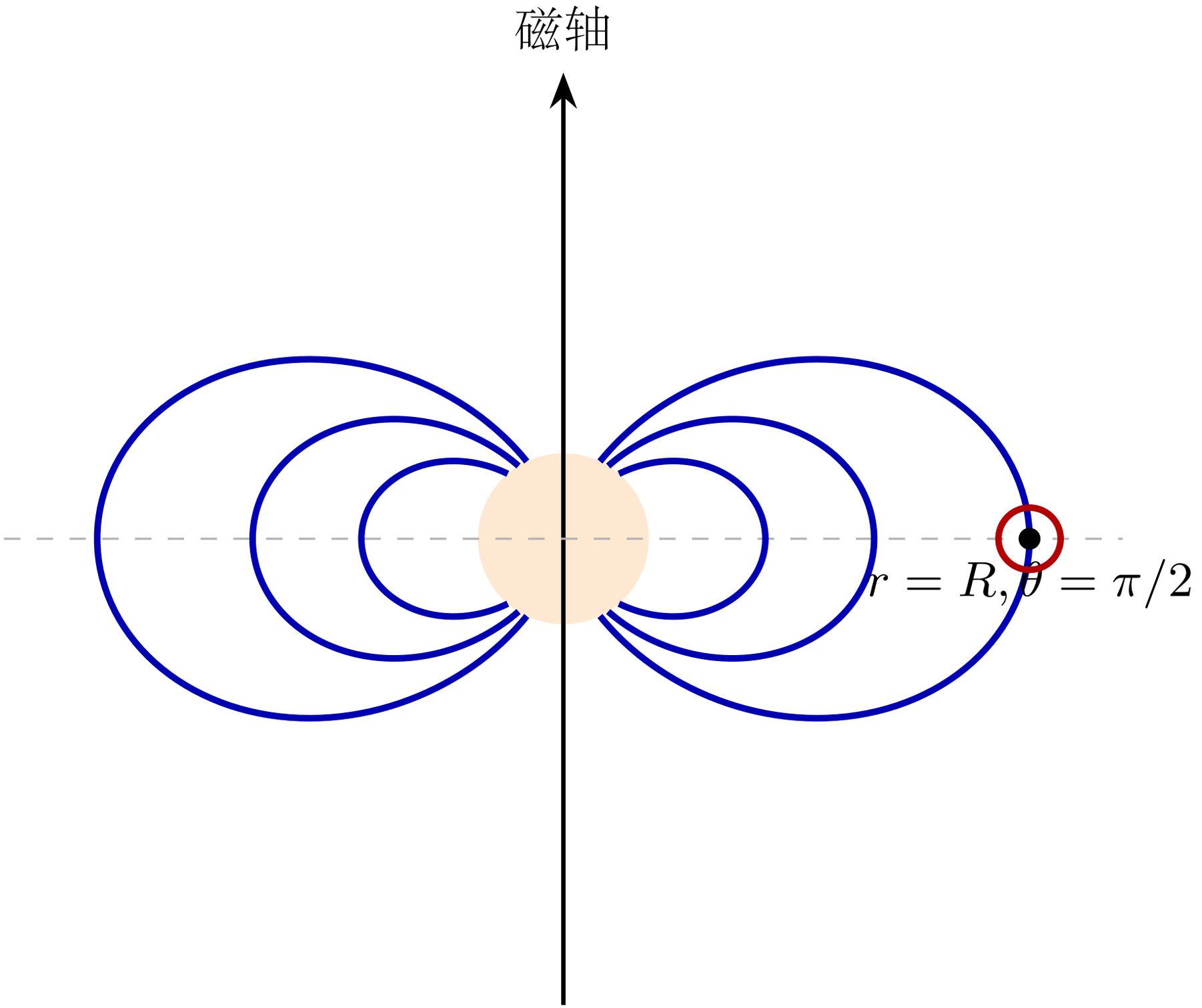
周期内部支路模型

此处 AC 为 r, C 串联



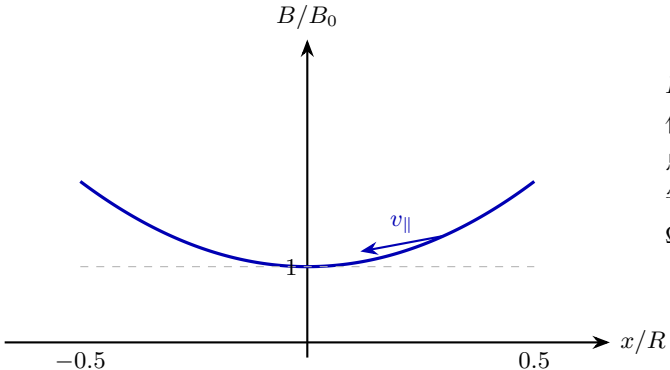
原图半无限导线端

边界另加入 $H\zeta^n$ 模



曲线按 $r = R_{\text{line}} \sin^2 \theta$ 绘制；地球大小仅示意

电子在选定磁力线赤道附近回旋并沿线缓慢运动



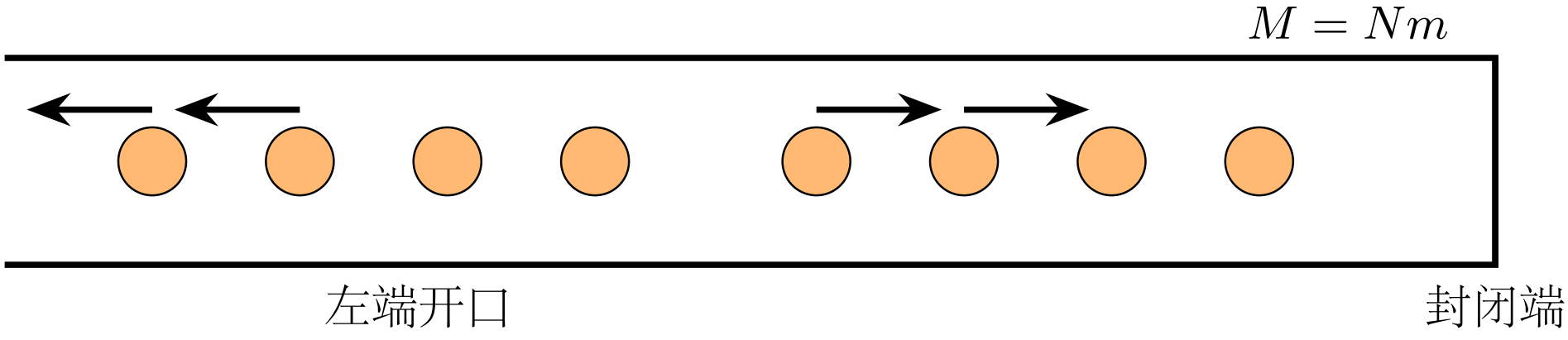
$B/B_0 = 1 + 9x^2/(2R^2) + O(x^4/R^4)$

仅赤道附近 $|x| \ll R$ 适用

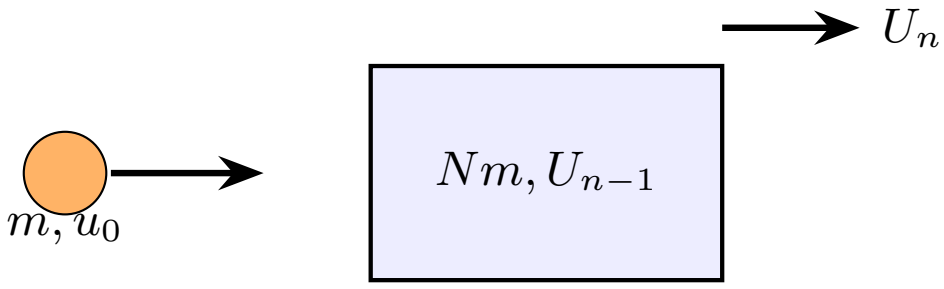
总能量、γ 不变

守恒量：$p_\perp^2/B$

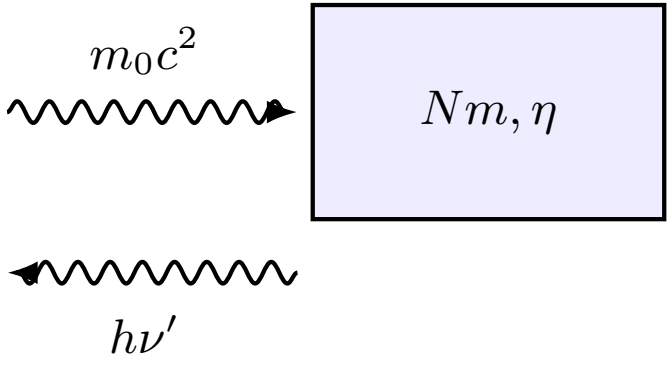
$\Omega = (3/\sqrt{2})a\omega/R$



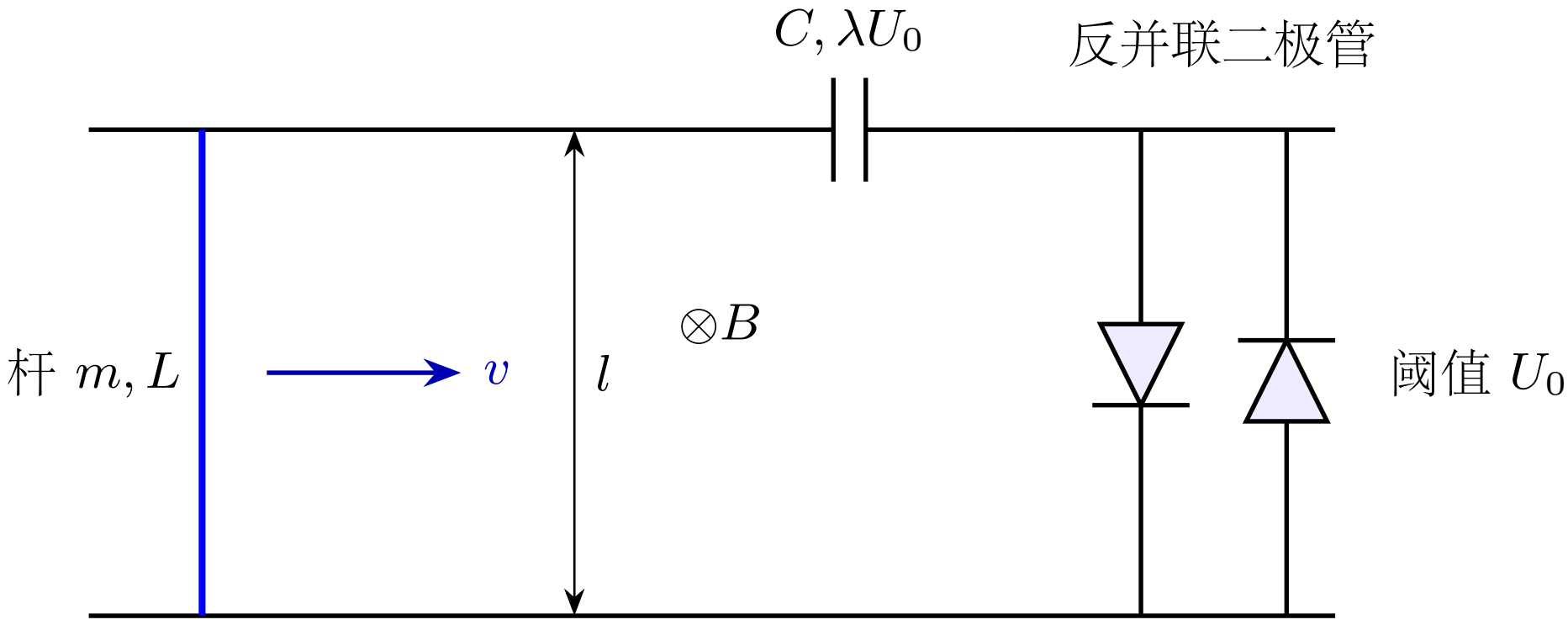
等速同质钢球在一维钢管内作完全弹性碰撞



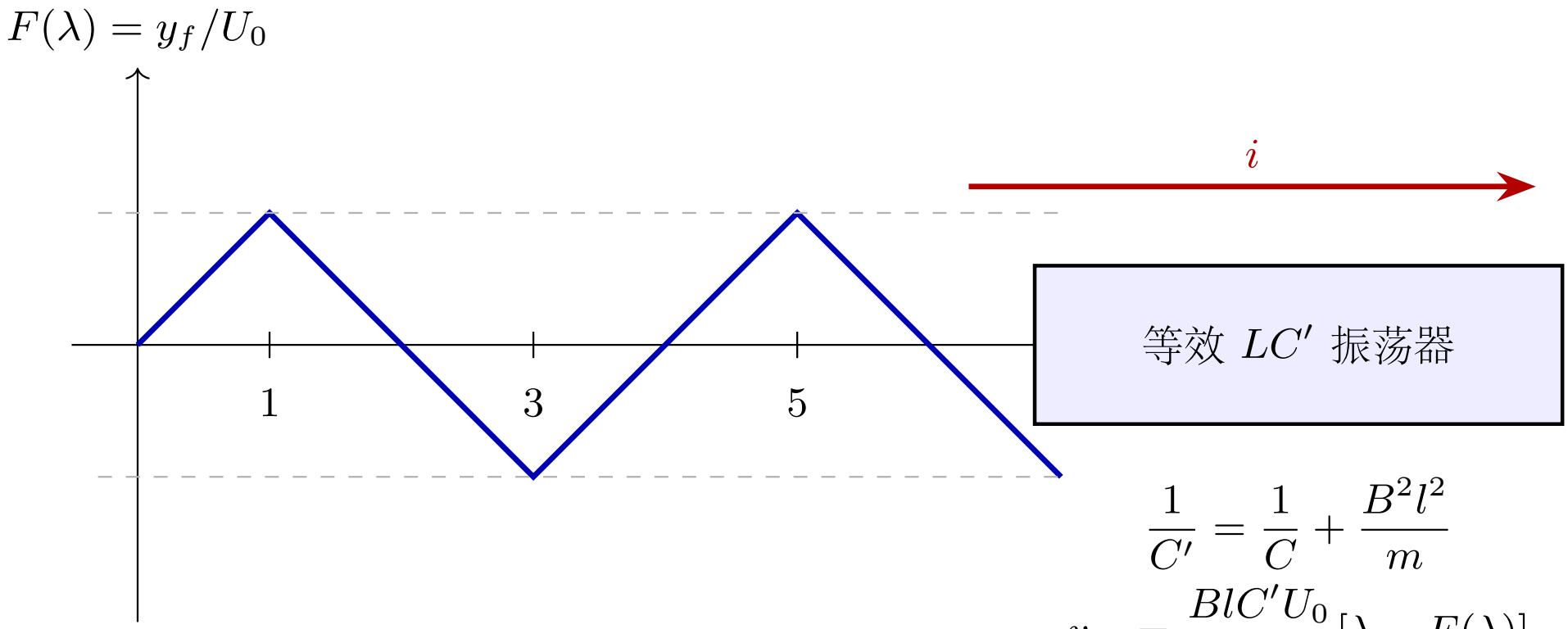
$$U_n = \frac{N-1}{N+1}U_{n-1} + \frac{2}{N+1}u_0$$



$$e^{\eta'} = e^{\eta} + \frac{2m_0}{Nm}$$



电容串入上导轨；二极管两端接上下导轨



$$F(\lambda + 4) = F(\lambda)$$

$$\frac{1}{C'} = \frac{1}{C} + \frac{B^2 l^2}{m}$$

$$v_\infty = \frac{BlC'U_0}{m}[\lambda - F(\lambda)]$$

二极管耗能 $U_0|i|$

F 周期，末速 v_f 非周期